Loan Delinquency & NPA Monitor: Project Summary

Excel project on public Prosper loan data.

1. Objective

Track the health of a loan portfolio the way a collections MIS team does: how many active loans are overdue, how late they are, how much money is at risk, and which borrower groups are the riskiest. The result is one dashboard a manager can read in 30 seconds and filter with slicers.

2. What we did (in short)

1. **Cleaned the data in Power Query.** Took a 113,937-row file down to 113,066 unique loans, kept 11 of 81 columns, filled blanks, and tagged every loan as Active, Closed or Written-off.
2. **Added helper columns with formulas** (IF, IFS, AND, YEAR) to sort active loans into days-past-due buckets, flag overdue and NPA loans, and group loans by size and year.
3. **Built 12 PivotTables** to measure delinquency by bucket, rating, loan size, state and year.
4. **Built the dashboard:** 5 KPI cards (text boxes linked to GETPIVOTDATA cells), 4 PivotCharts and 3 slicers connected to every pivot.

3. What we found

- **Portfolio:** 57,990 active loans worth $595.3M (original amount). 17,010 loans were written off, which is 15.0% of all loans.
- **Delinquency:** 2,057 active loans are overdue (3.55%). Of these, 320 are 90+ days late, which is the NPA group (0.55%).
- **Rating is the biggest driver:** delinquency rises from 0.71% for AA loans to 9.02% for HR loans, about 13 times higher.
- **Smaller loans are riskier:** 5.2% for loans under 5K against 1.6% for loans of 20K and above.
- **Write-offs follow the same pattern:** among finished loans, write-offs run from 4.7% (AA) to 38.2% (HR), so today's delinquency is an early warning.

- **Watch out:** 2014 loans look almost perfect (0.15%) only because they are new and have had little time to go overdue.

4. What to focus on further

- **The 1-30 day bucket (1,065 loans).** It is the biggest overdue group and the cheapest to recover.
- **The 31-60 and 61-90 day buckets (672 loans).** These are next in line to become NPAs.
- **D, E and HR ratings.** They have the highest delinquency and the highest eventual write-offs.
- **Small loans under 5K,** where delinquency is more than three times that of the largest loans.
- **High-volume states** (CA, NY, FL, TX, GA), which hold the most overdue loans by count.
- **Loans of the same age,** so older and newer groups can be compared fairly.

5. Suggestions

For the business

- Start collection calls and reminders at 1-30 days, before loans roll into the next bucket.
- Review underwriting and monitoring for D, E and HR borrowers and for small-ticket loans.
- Track overdue rates by loan age, not only by origination year.